

Mitomycin and fluorouracil (5FU) with radiotherapy

Mitomycin and fluorouracil (5FU) with radiotherapy is a cancer treatment. It is used to treat anal cancer and bladder cancer.

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What is mitomycin, fluorouracil (5FU) and radiotherapy?

Mitomycin and fluorouracil (5FU) with radiotherapy is a cancer treatment. It is used to treat [anal cancer](#) and [bladder cancer](#). It may sometimes be used to treat other cancers.

It is best to read this information with our general information about [cancer drug treatments](#), [pelvic radiotherapy](#), and the [type of cancer](#) you have.

Your cancer team will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

DPD testing before treatment

Before starting this treatment, you should have a blood test. This test checks for gene changes that affect how your body produces DPD. DPD is an enzyme that helps the body break down 5FU.

If you have low DPD levels or no DPD (DPD deficiency), you have more risk of serious or life-threatening side effects from 5FU. To reduce your risk, your doctor may give you:

- a lower dose of 5FU to start with
- a different type of chemotherapy.

Having low DPD levels does not usually cause any symptoms. Without a test, you will not know whether your DPD levels are low. Testing finds most people who are affected, but not all. Your doctor, nurse or pharmacist can give you more information about your risk of DPD deficiency, before you start treatment.

How mitomycin and fluorouracil (5FU) is given

You usually have mitomycin and fluorouracil (5FU) in a chemotherapy day unit or clinic as an outpatient or during a stay in hospital.

During your course of treatment, you will meet someone from your cancer team, such as a:

- cancer doctor
- chemotherapy nurse or specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

Before or on the day of each treatment, you will have a blood test. This is to check that it is safe for you to have treatment.

You will meet with a doctor, nurse or pharmacist before you have treatment. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your cancer drugs.

Your nurse will usually give you anti-sickness drugs before the cancer drugs. You will have your chemotherapy through 1 of the following:

- a [cannula](#) – a short, thin tube the nurse puts into a vein in your arm or hand
- a [central line](#) – a fine tube that goes under the skin of your chest and into a vein close by
- a [PICC line](#) – a fine tube that is put into a vein in your arm and goes up into a vein in your chest
- an [implantable port](#) (portacath) – a disc that is put under the skin on your chest or arm and goes into a vein in your chest.

Your cancer team may also give you anti-sickness drugs and other medicines to take home. Take all your capsules or tablets exactly as they tell you to.

Your course of chemoradiation

You will start the chemotherapy and radiotherapy on the same day.

Chemotherapy

On the first day of treatment, you will be given mitomycin over about 10 minutes and then you will be given 5FU.

The 5FU is usually given through a small pump over 4 or 5 days. Most people go home with the pump but some people may stay in hospital and have 5FU as a drip (infusion) instead.

If you are having 5FU treatment at home, you can wear the pump on your belt or carry it in a bag. Your chemotherapy nurse will explain how to look after the pump. They will also explain what to do if there is a problem.

Once the 5FU infusion is finished, you go to the chemotherapy day unit and a chemotherapy nurse will disconnect the pump. Sometimes a district nurse can do this at home.

You will have another 5FU infusion, usually during week 4 or 5 of radiotherapy treatment.

Radiotherapy

Once the 5FU infusion has started, you will have your first radiotherapy treatment.

You will have radiotherapy every day, Monday to Friday, for 5 to 7 weeks. How many treatment sessions you have will depend on the type of cancer.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

If your body does not produce enough of an enzyme called DPD, you may have a higher risk of severe or life-threatening side effects.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Side effects while treatment is being given

Some people may have side effects while they are having a cancer drug or shortly after they have it:

Allergic reaction

Some people have an allergic reaction while having this treatment. Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any of these signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

The drug leaks outside the vein

Sometimes cancer drugs that are given into a vein may leak outside the vein. If this happens, some drugs can damage the tissue near the vein. This is called extravasation. Extravasation is not common, but it is important that it is dealt with quickly. If you have any of the following symptoms around the vein during or after your treatment, tell your nurse straight away:

- stinging
- pain
- swelling
- if you have white skin, the area may become red
- if you have black or brown skin, the area may become darker.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an [infection](#). A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. It is important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

Your white blood cell count will usually return to normal before your next treatment. You will have a blood test before having more treatment. If your white blood cell count is low, your doctor may delay your treatment for a short time, until your cell count increases.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)

- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Diarrhoea

This treatment may cause severe [diarrhoea](#). Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. If you have a stoma, it will be more active than usual.

Your doctor, nurse or pharmacist may give you anti-diarrhoea drugs to take at home.

If you have diarrhoea or a mild increase in stoma activity, follow their advice about:

- taking anti-diarrhoea drugs
- how much and what type of fluids to drink
- any changes to your diet that might help.

Contact the hospital straight away on the 24-hour number if any of the following things happen:

- you have diarrhoea at night
- you have uncomfortable stomach cramps
- you have diarrhoea 4 or more times in a day
- you have a moderate increase in stoma activity
- the anti-diarrhoea drugs do not work within 24 hours.

You may need to give a stool sample, or go to hospital to have fluids through a drip or antibiotics.

Feeling sick

Your doctor, nurse or pharmacist will prescribe anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to, even if you do not feel sick. It is easier to prevent sickness than to treat it after it has started.

[If you feel sick](#), take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids. If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice. They may change your anti-sickness treatment. Let them know if you still feel sick.

Loss of appetite

This treatment can affect your appetite. Don't worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

Sore mouth and throat

This treatment may cause a [sore mouth and throat](#). You may also get mouth ulcers. This can make you more likely to get a mouth or throat infection. Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

Contact the hospital straight away on the 24-hour number, if:

- a sore mouth or throat affects how much you can drink or eat
- your mouth, tongue, throat or lips have any blisters, ulcers or white patches.

They can give you advice, and mouthwash or medicines to help with the pain or to treat any infection. Follow their advice and make sure you:

- drink plenty of fluids
- avoid alcohol and tobacco
- avoid food or drinks that irritate your mouth and throat.

Feeling tired

[Feeling tired](#) is a common side effect of this treatment. It is often worse towards the end of treatment and for some weeks after it ends. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

If you feel sleepy, do not drive or use machinery.

Hair loss

Your hair may get thinner. But you are unlikely to lose all the hair from your head. Hair loss usually starts after your first or second treatment. It is almost always temporary, and your hair will usually grow back after treatment ends.

Your nurse can talk to you about ways to cope with [hair loss](#).

Hand-foot (palmar-plantar) syndrome

This treatment can affect the palms of your hands and the soles of your feet. This is called palmar-plantar or hand-foot syndrome.

If you have white skin these areas may become red. If you have black or brown skin, these areas might get darker.

The skin on the palms of your hands and the soles of your feet may:

- be sore
- be painful, tingle, or swell
- peel, crack or blister.

If you have any of these symptoms, contact the hospital straight away on the 24-hour number. They can give you advice. This is especially important if you have any broken skin or if walking is difficult. They can prescribe creams and painkillers to help.

You can care for your hands and feet by:

- keeping your hands and feet cool by washing in cool water
- gently moisturising your hands and feet regularly
- wearing gloves to protect your hands and nails when working in the house or garden
- wearing loose cotton socks and avoiding tight-fitting shoes and gloves.

Effects on the heart

5FU can affect how the heart works. You may have [tests](#) to check how well your heart is working. These may be done before, during and after treatment.

You may develop [heart problems](#) even if these test results are normal. Very rarely, 5FU causes heart failure or a heart attack. The risk of this happening is very low. It happens to less than 1 in 100 (1%) of people who have 5FU. But it is important that you know about it.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.

Always call 999 if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Skin changes

Chemotherapy can affect your skin and nails. If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

This treatment can also:

- cause a rash, which may be itchy
- make your skin darker in some areas
- make any area treated with radiotherapy become red or sore. If you have white skin the area will become red and if you have black or brown skin the area might become darker.
- make you more sensitive to the sun.

Your skin may burn more easily during treatment and for several months after. Use a sun cream of at least SPF 30. SPF stands for sun protection factor. Cover up with clothing and wear a hat.

Always tell your doctor, nurse or pharmacist about any [skin changes](#). They can give you advice or prescribe creams or medicines to help. Changes to your skin and nails are usually temporary and improve when treatment ends.

Effects on the eyes

This treatment can affect your eyes. Your eyes may become watery and feel sore. Your doctor, nurse or pharmacist can give you advice and eye drops to help with this.

Contact the hospital on the 24-hour number if:

- your eyes get red and inflamed (conjunctivitis)
- you have pain in your eyes
- your eyes are sensitive to light
- you notice any change in your vision.

Effects on the liver and kidneys

This treatment can affect how your kidneys or liver work. This is usually mild and goes back to normal after treatment ends. You will have blood tests to check how well your kidneys and liver are working.

Contact the hospital on the 24-hour number if you:

- have blood in your urine (pee)
- are passing less urine or peeing less often than usual.

Drinking fluids also helps protect your kidneys. The advice is usually to try to drink at least 2 litres (3½ pints) of fluid each day. But follow any advice from your doctor, nurse or pharmacist about how much is right for you.

Rarely mitomycin can cause a kidney problem called haemolytic uraemic syndrome (HUS). This is very rare but it can be serious. Tell your doctor if you feel tired and breathless, or if you have any bruising or bleeding you cannot explain.

Effects on the lungs

This treatment can cause changes to the lungs. Tell your doctor, nurse or pharmacist if you develop:

- a cough that does not go away
- wheezing
- breathlessness.

You should also tell them if any existing breathing problems get worse. You may have tests to check your lungs.

Difficulty breathing

5FU can cause sudden tightness (spasm) in the muscles in your chest for a short time. Contact the hospital straight away on the 24-hour number if you have any of these symptoms during treatment:

- breathlessness
- a cough that does not go away
- wheezing.

Always call 999 if you have difficulty breathing.

Mood changes

Sometimes 5FU can cause mood changes such as feeling happy, excitable or agitated. Tell your doctor, nurse or pharmacist if you notice this.

Effects on the brain

Rarely, this treatment causes a brain condition that can be serious. You can make a full recovery from this. But it must be diagnosed and treated quickly.

This condition can cause:

- a headache that does not get better
- drowsiness or confusion
- problems with balance, movement or speaking
- changes in eyesight
- fits (seizures).

If you have any of these symptoms, it is important to either:

- contact the hospital straight away on the 24-hour number
- go to the hospital straight away.

You should not drive yourself to hospital.

Other information

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix[®] for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Sex

It is possible that small amounts of chemotherapy may be passed on through vaginal fluids or semen. If you have sex in the first few days after treatment, your cancer team will usually advise using condoms or a dental dam to protect your partner.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Date reviewed

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help